

UNIVERSITY OF TRANSPORT
HO CHI MINH CITY



CAPSTONE PROJECT
FACULTY OF INFORMATION
TECHNOLOGY

TOPIC

DEVELOPING A HORSE RACING
MANAGEMENT SYSTEM

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Best regards,
[Group 3].

Definition and Acronyms

Acronym	Definition
API	Application Programming Interface
BA	Business Analysis
BR	Business Rule
DB	Database
ERD	Entity Relationship Diagram
GUI	Graphical User Interface
MVP	Minimum Viable Product
PM	Project Manager
SDD	Software Design Description
SPMP	Software Project Management Plan
SRS	Software Requirement Specification
UAT	User Acceptance Test
UC	Use Case
UI	User Interface
UX	User Experience

1 I. Project Introduction

1.1 1. Overview

1.1.1 1.1 Project Information

- **Project name:** Horse Racing Tournament Management System
- **Project code:** [Project Code]
- **Group name:** [Group 3]
- **Software Type:** Web application

1.1.2 1.2 Project Team

Full Name	Role	Email	Student ID
	Supervisor		
	Leader		
	Member		
	Member		
	Member		
	Member		
	Member		

1.2 2. Product Background

Horse racing is a competitive sport that involves multiple stakeholders — including race organizers, horse owners, jockeys, referees, and spectators — each requiring timely and accurate information to participate effectively. Traditionally, the management of horse racing tournaments has been carried out through manual processes, which are error-prone, time-consuming, and difficult to scale.

The *Horse Racing Tournament Management System* is proposed to address these challenges by providing a centralized, digital platform that automates and streamlines the entire tournament lifecycle: from user registration and horse enrollment, through race scheduling and referee assignment, to result recording and leaderboard updates.

1.3 3. Existing Systems

1.3.1 3.1 Manual / Spreadsheet-Based Management

Description: Many racing organizations currently rely on spreadsheets (Microsoft Excel, Google Sheets) and printed forms to track registrations, race schedules, and results.

Drawbacks:

- No real-time synchronization — updates must be manually distributed to all parties.
- High risk of data entry errors and version conflicts.
- No role-based access control; sensitive data is exposed to unauthorized personnel.
- Difficulty generating automated leaderboards or performance statistics.

1.3.2 3.2 Generic Sports Management Platforms

Description: General-purpose sports management tools (e.g., SportyHQ, TeamSnap) provide basic scheduling and communication features for various sports.

Drawbacks:

- Not designed for horse racing — lack domain-specific features such as horse profile management, jockey assignment, and race-specific refereeing workflows.
- Limited support for multi-role workflows (Horse Owner, Jockey, Race Referee, Spectator).
- Integration with race result hardware or official timing systems is typically not available.
- High licensing costs with features irrelevant to horse competitions.

1.4 4. Business Opportunity

By centralizing the management of horse racing tournaments, the system enhances collaboration, improves communication among all stakeholders, and optimizes the workflow from race registration to result publication. The key business opportunities include:

- **Efficiency gains:** Automating registration, scheduling, and result processing reduces administrative workload by an estimated 60%.

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- **Data accuracy:** A single source of truth eliminates duplication and ensures consistent, up-to-date information across all roles.
 - **Fan engagement:** Real-time leaderboard and race schedule visibility enhances the spectator experience and can increase event attendance.
 - **Scalability:** The platform can grow to support multiple concurrent tournaments and additional user roles without significant re-engineering.
 - **Auditability:** A complete digital trail of all registrations, race results, and administrative actions supports compliance and dispute resolution.

1.5 5. Software Product Vision

For racing organizations and equestrian clubs **who** need a reliable and transparent platform to organize horse racing tournaments, the *Horse Racing Tournament Management System* is a web application **that** provides end-to-end tournament management — from horse and jockey registration to real-time race results and leaderboard updates. **Unlike** manual spreadsheet-based processes or generic sports platforms, **our product** delivers a purpose-built, role-aware solution that serves Admins, Horse Owners, Jockeys, Race Referees, and Spectators within a single integrated system.

1.6 6. Project Scope & Limitations

1.6.1 6.1 Major Features

6.1.1 Major Features for Admin

- FE-01: Login / Logout
- FE-02: Manage user accounts (create, update, deactivate)
- FE-03: Manage tournament information
- FE-04: Manage race schedule and race lanes
- FE-05: Assign referees to races
- FE-06: Publish results and update leaderboard
- FE-07: Send system notifications

6.1.2 Major Features for Horse Owner

- FE-01: Login / Logout
- FE-02: Register and manage horse profiles
- FE-03: Enroll horses in tournaments
- FE-04: View race schedule and results for owned horses
- FE-05: View leaderboard standings

6.1.3 Major Features for Jockey

- FE-01: Login / Logout
- FE-02: View and accept race invitations
- FE-03: View assigned race schedule
- FE-04: View personal performance history and rankings

6.1.4 Major Features for Race Referee

- FE-01: Login / Logout
- FE-02: View assigned race list
- FE-03: Enter and confirm race results
- FE-04: Flag incidents or disqualifications

6.1.5 Major Features for Spectator

- FE-01: Register / Login (optional)
- FE-02: View race schedule and tournament information
- FE-03: View real-time results and leaderboard
- FE-04: View horse and jockey profiles

1.6.2 6.2 Limitations & Exclusions

- **LI-1: Live timing hardware integration** — The current version does not integrate with physical race timing sensors or photo-finish equipment; results are entered manually by the Race Referee.
- **LI-2: Payment gateway** — Online betting, entry fee payments, and prize disbursements are out of scope for this release.
- **LI-3: Mobile native application** — Only a responsive web application is delivered; dedicated iOS/Android apps are not included.
- **LI-4: Third-party horse registry** — Integration with official national horse registries or stud-book databases is not supported in this version.
- **LI-5: AI/ML prediction features** — Automated race outcome prediction or ranking prediction based on historical data is deferred to a future release.

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- 2 II. Project Management Plan**
 - 3 III. Software Requirement Specification**
 - 4 IV. Software Design Description**
 - 5 V. Software Testing Documentation**
 - 6 VI. Release Package & User Guides**